

11	C	$g = \frac{GM}{r^2}$ $g \propto \frac{M}{r^2}$ Let the quantities with subscript 'J' represent that of Jupiter while those with subscript 'E' represent that of Earth. $\frac{g_J}{g_E} = \frac{M_J}{M_E} \left(\frac{r_E^2}{r_J^2} \right)$ $\frac{g_J}{g_E} = (318) \left(\frac{6370 \times 10^3}{7.15 \times 10^7} \right)^2$ Since $g_E = 9.81 \text{ N kg}^{-1}$, $g_J = (318) \left(\frac{6370 \times 10^3}{7.15 \times 10^7} \right)^2 (9.81)$ $= 24.8 \text{ N kg}^{-1}$
12	B	$\Delta U = m \Delta \phi \propto \Delta \phi$